Student Performance Analysis

By Cornelius Enetomhe

Educational Dashboard

Executive Summary

Overview

This educational dashboard provides a comprehensive view of student performance across years, subjects, assessments, teachers, and demographics. This analysis offers a comprehensive overview of student academic trends, identifying key areas of strength and opportunities for improvement. We'll explore performance across subjects, assessments, and demographics to inform strategic educational enhancements.

Overall results indicate strong academic performance, improved pass rates, and consistent teaching outcomes, even as student enrollment increased significantly.

Key Performance Highlights

Improved Pass Rates
From 69.2% (2022) to a stable 80% (2023-2025).

Consistent Scores
Average student scores maintained academic standards.

Targeted Support Needed
Lower averages in P.E., Foreign Language, Computer Science.

Project-Based Success
Assessments yielded best student performance.

Mid-Year Dips
June-July performance dips, strong year-end recovery.

Manageable Failure Rate
Overall, 20%, requires attention with enrollment growth.

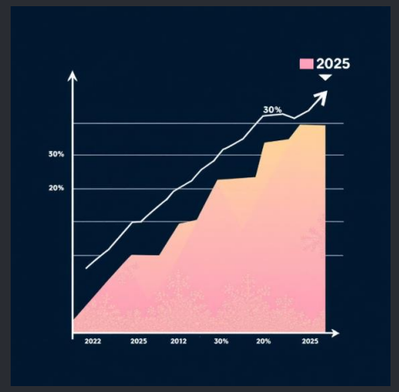
Consistent Teachers
Minimal variation in teacher performance.

Gender Performance Gap
Female students outperform males by 3.78 points.

Key Insights: Progress & Stability

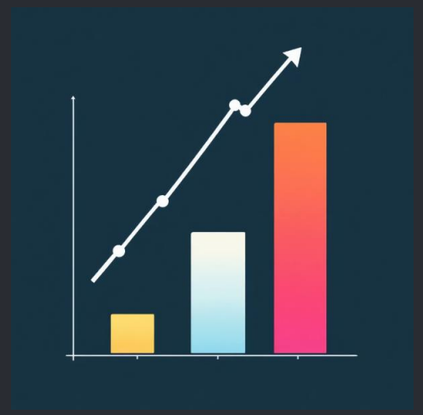
Pass Rate Soars with Enrollment Growth

Student pass rates improved by 10.8% (69.2% in 2022 to 80% in 2023-2025), even as student population grew from 13 to 50.



Decreasing Failure Rate

Failure rate decreased from 30% to 20% of the student population from 2022 to 2025, demonstrating positive trend.

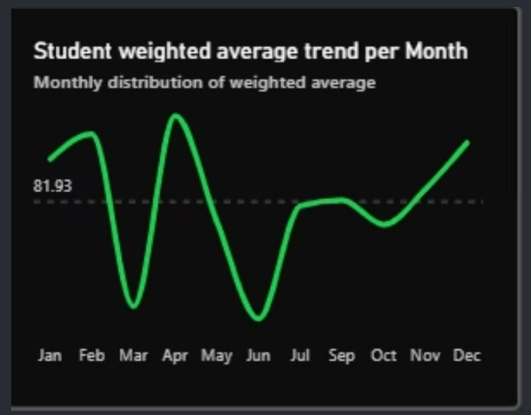


Consistent Academic Standards

Average student scores across subjects remain around the 70% mark, indicating stable academic quality.

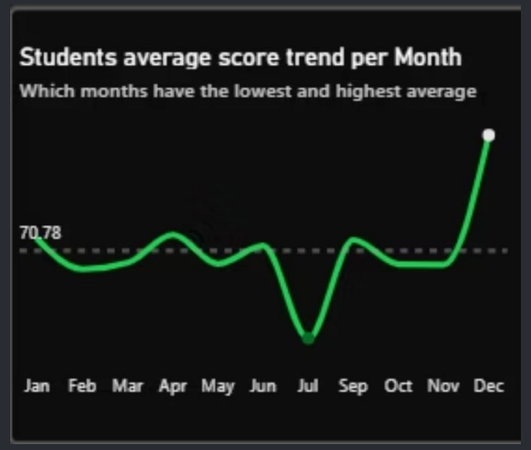
These trends highlight effective educational strategies adapting to significant growth.

Key Insights: Monthly Performance Trends



Mid-Year Performance Dips

Performance consistently declines during June–July, followed by strong recovery towards year-end.



Periods Below Average

Majority of months show student averages below the mean, with lows in March, June, and July.

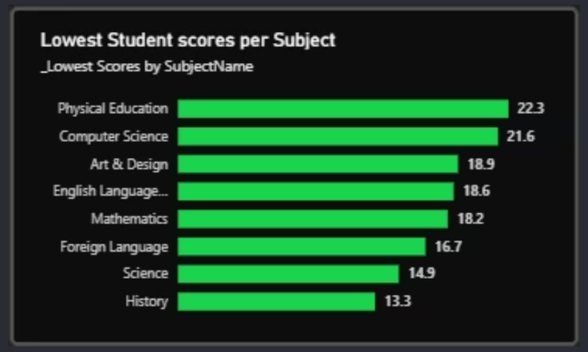
Peak Performance

Best performance in January, April, November, and December, with Q4 as the strongest quarter.

Strategic Intervention

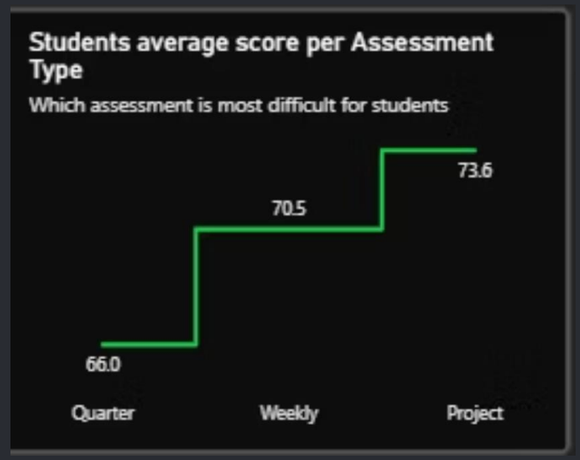
Identifying these patterns allows for targeted interventions to bolster performance during weaker periods.

Key Insights: Assessment & Subject Performance



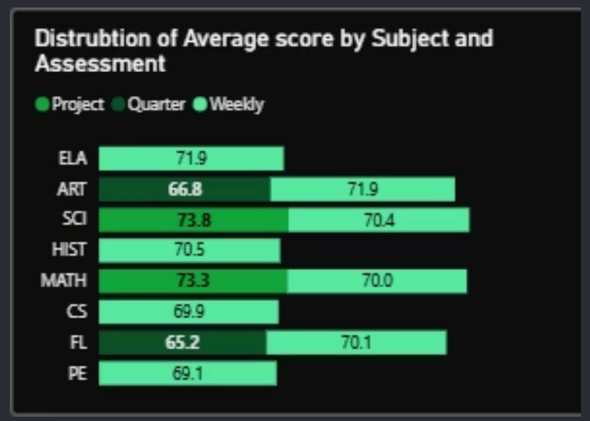
Project-Based Success

Project-based assessments yielded highest average scores (73.6%), favoring applied learning.



Quarterly Challenges

Quarterly assessments recorded lowest averages (66.0%), posing the most difficulty for students.



Subjects Needing Support

P.E., Foreign Language, and Computer Science show lower averages, indicating areas for targeted support.

Strongest Subjects

English Language, Art & Design, Science, and History emerge as top-performing subjects.

Insights for Curriculum

These findings inform curriculum adjustments and pedagogical approaches.

Key Findings: Scalability & Subject Focus

Performance Scaled Successfully with Enrollment Growth

- Student enrollment **increased nearly 4x** (13 → 50 students) between 2022 and 2025.
- Pass rates **improved from 69.2% to a sustained 80%**, indicating effective instructional quality.
- Average scores across subjects remained stable at **~70%**.

Implication: The system is resilient and capable of supporting further growth without immediate risk to outcomes.

Subject Performance Concentrated Around a Narrow Band

- **Top-performing subjects:** English Language, Art & Design, Science, and History.
- **Underperforming subjects:** Physical Education, Foreign Language, and Computer Science.
- Performance variance is narrow (**≈3 points**), indicating systemic rather than isolated issues.

Implication: Targeted, subject-specific interventions can yield quick performance gains.

Key Findings: Assessment & Workload

Assessment Design is a Key Performance Driver

- **Project-based assessments** deliver strongest outcomes (avg. 73.6%).
- **Quarterly assessments** show weakest performance (avg. 66.0%).
- Weekly assessments sit mid-range, suggesting assessment frequency correlates with performance.

Implication: Students perform better in applied and continuous evaluation. High-stakes assessments are primary risk points.

Mid-Year Performance Dip Signals Structural Fatigue

- Performance declines consistently during **June–July**, followed by strong year-end recovery.
- Pattern indicates **academic load concentration** rather than capability gaps.

Implication: Redistributing workload or introducing mid-year academic support could materially improve outcomes.

Key Findings: Teaching & Demographics

Consistent Teaching Quality

Minimal variance in average scores across teachers. No outlier instructors driving poor or exceptional performance.

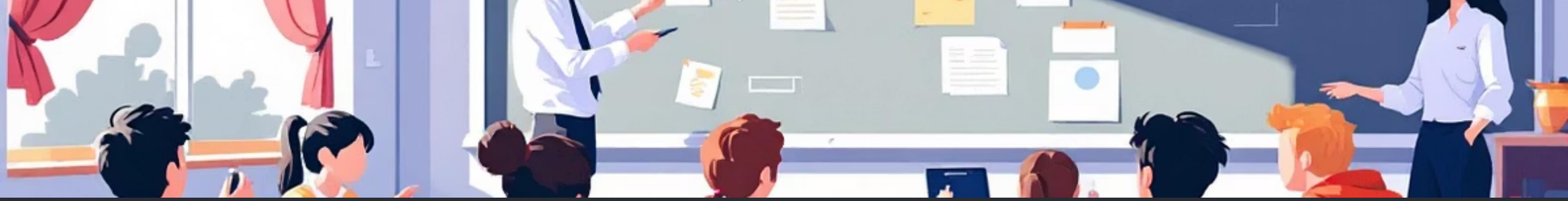
Implication: Performance issues are **systemic**, not personnel-driven. Solutions should focus on curriculum and assessment design.

Manageable Demographic Gap

Female students outperform male students by **3.78 points**.

Gap is consistent but not widening.

Implication: While not critical, targeted engagement strategies for male students could close the gap over time.



Recommendations for Improvement

Redesign Quarterly Assessment Preparation

Introduce structured revision cycles and formative checkpoints to improve performance in challenging assessments.

Prioritize Support for Lower-Performing Subjects

Focus on Physical Education, Foreign Languages, and Computer Science with targeted academic interventions.

Address Mid-Year Performance Dip

Implement academic recovery programs or workload smoothing strategies to maintain consistent performance throughout the year.

Maintain Current Teaching Model

No immediate intervention required at the staff level, as teaching quality is consistently high.